Hibbeler Ch. 8: Dry Friction

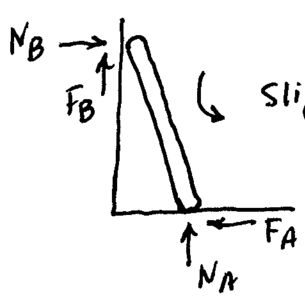
"Fast Fundamental Facts about Friction"

1. Dry friction exists between dry surfaces.
   Fluid friction (lubricated surface) behaves diffently, it's a fluid mechanics problem. ME's learn about it in Machine Design.

2. Friction is both a blessing and a curse.
   Blessing: Enables us to walk, drive a car, play sports
   Curse: Robs energy from machines, produces heat, wear

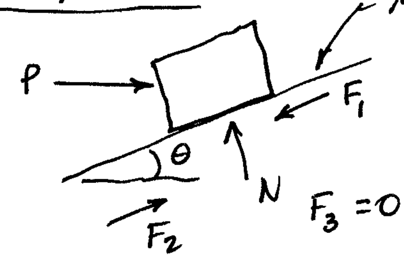
3. Friction forces develop to oppose relative motion.

Ladder Problem



Crate on slope.
Two modes possible.

Coeff of Friction μ

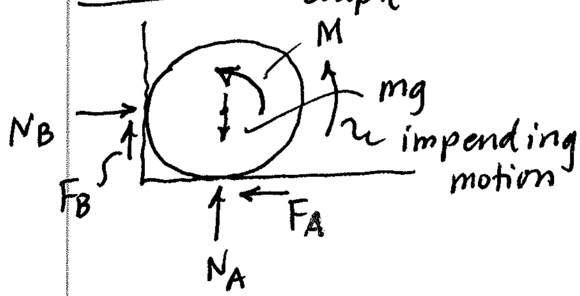


For large P, motion impends upslope, F_1 opposes it.

For small P (and small μ), motion may impend down slope, F_2 opposes it.

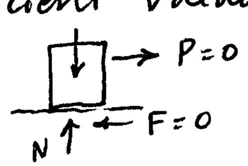
There is a P at which F = 0.

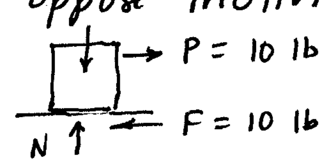
Spool



★ Before draw FBD, think about "What direction will motion impend" at point(s) of contact.
Then draw friction forces to oppose this motion.

4. Friction forces are reactive forces; they develop sufficient values to oppose motion.

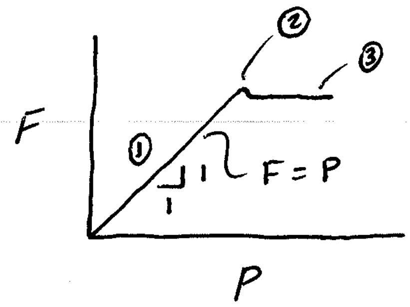


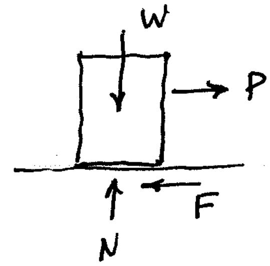


If it can, F exactly = P ...

5. However, friction is an <u>inequality</u>. $\boxed{F \le \mu N}$

Once F reaches its limiting value, motion "impends" (on verge of moving) or occurs. Moves!





Two coefficients of friction:

μ_s -- "static friction" (this class) } Usually $\mu_s > \mu_k$

μ_k -- "kinetic friction" (a dynamics problem, motion occurs)

$P - F = ma$

[See ①, ②, ③ on diagram above]

<u>Cases</u>

① $F < \mu_s N$. Slipping not impending.
$F =$ Static friction force needed to maintain equilibrium.

② $F_s = \mu_s N$. Slipping impending.
$F_s =$ limiting Static force

③ $F_k = \mu_k N$. Slipping (motion) occurring.
A dynamics problem. $P - F = ma$

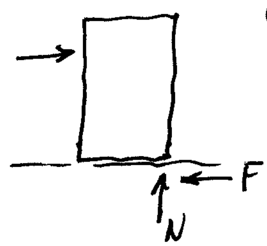
6. Important! Read problem carefully. Do not automatically set $F = \mu N$.
Some problems solve for F_A, N_A, etc. from Equil Eqns, then check $\mu_s \le (F_A / N_A)$ ...

7. Look for the phrase "motion impends" or "impending motion" in the problem statement. This is a clue that you can set $F = \mu N$. May also have to check tipping or other modes of motion.

Learn to look for modes of motion; see example problems and HW Helps Videos.

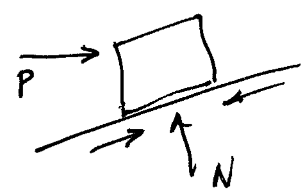
8. Possible multi-mode problems:

HW Prob



(a) Crate tip-slip. Check both tip and slip.

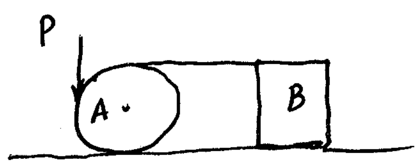
(b)



Crate can slip up or down slope.

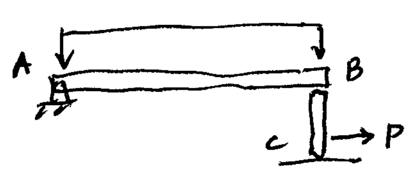
HW Prob

(c)



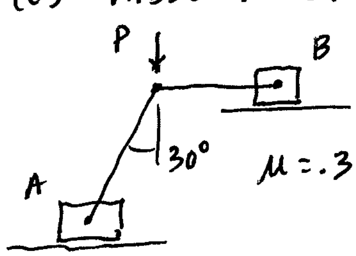
- Wheel A can slip?
- Block B can tip?
- Block B can slip?

(d) Hibbeler Ex Prob 8.4



Post BC can slip at B or C first. Must check both possibilities.

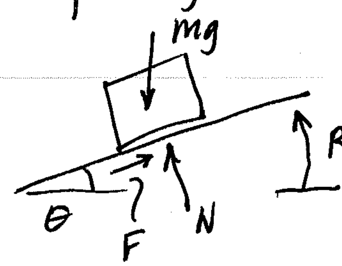
(e) Hibbeler Ex Prob 8.5



Either block A or B can move first. Must check both cases.

9. "Angle of Friction" (aka "Angle of Repose")
(See Hibbeler Ex Problem 8.2
Vending Machine tip/slip out of truck)

Easy way to measure coeff of friction:



- Place object on surface.
- Raise surface until object slides...

This θ is "Angle of Friction"

$+\nearrow \quad F = mg\sin\theta$

$\nwarrow \quad N = mg\cos\theta$

$$\mu = \frac{F}{N} = \frac{mg\sin\theta}{mg\cos\theta} = \tan\theta$$

$$\boxed{\mu = \tan\theta}$$

See examples in class.

Yardstick + various objects...